GraphOne / FrontierAtlas - Technical Architecture

Global AI Intelligence Graph Ingestion Engine | Production Design Document

1. Scale Strategy (Ingesting 500,000+ Entities Hand-Free)

To acquire 500,000+ Startups, Products, Research Papers, Jobs, and News items without manual intervention, our ingestion architecture decouples discovery, extraction, enrichment, and storage into a distributed event-driven actor system. Built on Playwright Async, aiohttp, Celery, and Kafka over Kubernetes, the system handles hundreds of thousands of requests concurrently.

Core Scale Mechanisms:

- **Sharded Distributed Workers:** Web crawlers run as async microservices. Crawl targets are sharded dynamically via consistent hashing across cluster pods.
- **Anti-Bot Proxy & Stealth Swarm:** Rotates proxy IPs (Smartproxy/BrightData) and spoofing browser TLS stack fingerprints to bypass Cloudflare and Datadome protections.
- **Recursive Seed Discovery:** Automated crawlers continually discover new entities from arXiv feeds, YC directories, and ProductHunt releases.

2. Managing LLM 413 Context Windows & 429 Rate Limits

LLM payload processing faces two main bottlenecks: **413 Payload Too Large** (context overflows) and **429 Rate Limits**. Our multi-tier engine guarantees resilient structuring through payload chunking and failover chains.

- **413 Payload Chunking:** Raw HTML is stripped of non-content tags (<script>, <style>). Large payloads are chunked into semantically dense sections (<3,500 tokens) retaining structural header anchors.
- **Multi-Tier LLM Fallback Chain:** Primary: Gemini 1.5/2.0 Flash → Secondary: Groq Llama 3 70B → Tertiary: DeepSeek Chat / Rule-based Extractor.
- **429 Exponential Backoff with Jitter:** Token bucket rate-limiter delays requests using *Wait = min(T_max, T_base * 2^attempt + jitter)*, coordinated across pods via Redis.

3. Distributed Freshness Tracking (< 24 Hours)

High-velocity signals (News & Jobs) must be guaranteed to have been published within the last 24 hours without duplicate processing across crawler nodes:

- **Redis Bloom Filters:** O(1) timestamp check skips URLs processed within the last 24 hours.
- **Content SHA-256 Hash Deduplication:** SHA-256 hashes of normalized article texts merge syndicated press releases into a single event.
- **Date Normalization Engine:** Parses relative dates ('2 hours ago'), RSS RFC-822, and ISO strings, falling back to DOM modification heuristics if metadata is missing.

4. Storage Strategy (Primary DB, Graph, Vector Storage)

We deploy a Tri-Storage Hybrid Architecture to power complex intelligence graph queries:

Database	Technology	Role & Justification
Primary DB	PostgreSQL 16 + TimescaleDB	ACID compliant relational storage for canonical records, time-series telemetry, and schema validation.

Knowledge Graph	Neo4j Enterprise	Maps relationships: (Founder)-[:FOUNDED]->(Startup)-[:PRODUCES]->(Product) for sub-ms multi-hop traversals.
Vector DB	Qdrant / Milvus	Stores text-embedding-3-large vectors for semantic search, cross-entity matching, and RAG discovery.